

Project Proposal

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1 General Problem

Users in multi-turn dialogues frequently change subjects or return to previous ones, particularly when explaining complex topics (e.g. explaining the Transformer architecture) with various components (e.g. tokenization, attention mechanism). Current LLM-based agents tend to get sidetracked by irrelevant details. This creates significant challenges for Large Language Models (LLMs) in tracking conversational context and maintaining coherent interactions. The difficulty comes from the models' tendency to either incorrectly carry over irrelevant context from previous discussions or fail to recall pertinent information when a topic is resumed (Li et al., 2025).

Wang and Sun (2025) used the proactive interference (PI) analogy to describe this problem. In cognitive science, proactive interference refers to the difficulty in recalling new information due to interference from previously learned information. It's attributed to the brain's inability to actively unbind outdated associations from working memory before encoding new information. For instance, after updating a password that hasn't been changed for years, people might repeatedly type the old password by mistake because the outdated association interferes with our ability to recall and use the new one. Current LLMs lack the ability to deliberately forget information and select the right context for responses. Simply using instructions in the prompt to trigger forgetting hasn't fully solved this issue, which means current architectures can't adequately address these shortcomings.

2 Related Work

Hwang et al. (2024) introduced MP2D, a method that maps dialogue topic flow using knowledge graph entity relationships. While MP2D effectively reflects human conversational dynamics, its focus on topic switch detection limits its utility for evaluating context comprehension. To assess the im-

pact of proactive interference (PI) in real-world conversations, a dedicated QA dataset with topic switches is necessary. Xie et al. (2021) and Bai et al. (2024) offer fine-grained dialogue data with topic transitions. However, both datasets are designed for response generation, which is unsuitable for our objective. The open-ended nature of their responses hinders the quantitative measurement of PI's effects; therefore, we prioritize QA data with closed-ended questions.

While Wang and Sun (2025) introduces a valuable framework for understanding proactive interference in LLMs, their proposed metric primarily focuses on synthetic, controlled settings, such as sequential key-value updates. This approach, though insightful for isolating interference, does not fully capture the complexities of natural multi-turn dialogues where topics shift organically and prior information's relevance dynamically changes. Therefore, to effectively quantify the impact of proactive interference in real-world conversational scenarios, a modified or new measurement specifically designed for the nuanced nature of natural dialogue is essential for further exploration.

3 Approach

Due to the lack of high quality conversational data including context-switch with QA pairs, we first generate a synthetic data from combining existing conversations. Then we leverage graph-based agents to retrieve the related information, and use that to contextualize LLM to generate related responses.

3.1 Data Formulation

To simulate natural topic switching, we flatten multiple conversation threads into a single, continuous dialogue. This approach is beneficial because it forces the model to manage and prioritize interleaved topics, mirroring the cognitive load humans experience in complex multi-turn conversations. Moreover, it aligns with practical user interaction,

allowing seamless communication without manual session management, reflecting ideal real-world usage. Specifically, we first split the dialogue into chunks with normal length of a conversation and then concatenate the chunks from one topic after another. For that we'd use LongMemEval, a dataset providing dialogue histories in the form of multiple haystacks and corresponding QA pairs, which is aligned with our design. Additionally, LongMemEval labels the QA pairs with five core long-term memory abilities, thus we can look into the effect of PI on different abilities for more insight.

3.2 Potential Solution

To eliminate PI, an agent should be able to identify the entities in the conversation and distinguish which entities are suitable to compose an immediate response. Since naive prompts have a limited effect, a curated planning pipeline should be included for better judgment. Li et al. (2024) proposed GraphReader, a graph-based agent system designed to significantly enhance LLMs' long-context capabilities by structuring extensive texts into a knowledge graph. This system constructs a graph from documents by extracting atomic facts and key elements, which are then navigated by an LLM-powered agent through rational planning and strategic node exploration. We plan to leverage GraphReader's outstanding ability to identify crucial information and add prompts for unbinding undesired nodes, which can be the solution to PI.

4 Hypotheses Verification

Wang and Sun (2025) proposed Interference Endurance Score (IES) to quantify a model's ability to resist PI. The IES is defined as the area under the curve (AUC) of retrieval accuracy, calculated across log-scaled update counts, i.e., number of key updates in a hashmap saving the data. Figure 1 show comparison between two models using this measurement. We can adapt the IES by replacing the log-scaled number of repetitions with the number of dialogue turns. Also, replacing the denominator with other statistics that may be related to difficulty will be measured, e.g. number of entities.

5 Data Access & Existing Code

LongMemEval can be easily access through [github](#). Prompts for knowledge extraction and planning can be found in the appendix of Li et al. (2024).

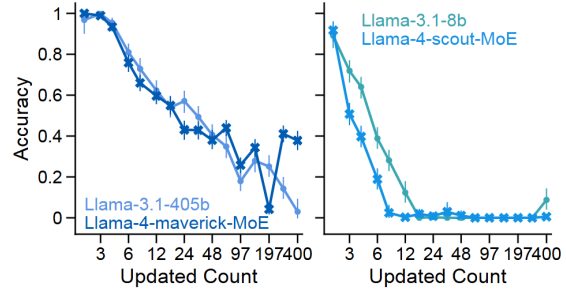


Figure 1: Comparison of retrieval accuracy between Mixture-ofExperts (MoE) and dense models from Wang and Sun (2025). The IES is defined as the area under the curve (AUC) of retrieval accuracy, calculated across log-scaled update counts.

Topic Shift	
Case:	Evaluation:
Human: I want to watch a film with a complex plot. Any suggestions? Assistant: For a film with a complex plot, I'd recommend 'Inception' directed by Christopher Nolan. It's a thrilling science fiction heist movie that delves into the realms of dream manipulation and features several layers of narrative complexity. Human: What's a healthy breakfast option for someone on a keto diet? Assistant: For someone on a keto diet, a healthy breakfast option could include an omelet with spinach, cheese, and avocado, which is high in healthy fats and fiber while being low in carbs. Another option is a smoothie with almond milk, spinach, peanut butter, and chia seeds. Human: Does the movie leave the ending open to the audience's interpretation? Assistant: The movie 'Leave' doesn't exist. Could you please provide more context or clarify the title of the movie you are referring to?	The AI assistant's response in the last round of conversation fails to acknowledge the previous reference to the movie 'Inception' and incorrectly assumes a misunderstanding regarding the existence of the movie title 'Leave'. This is a significant error as the human was not asking about a movie called 'Leave' but inquiring about the ending of 'Inception', which was discussed earlier in the conversation. The assistant's response does not detect and adapt to the continuation of the initial topic about the movie 'Inception'. Instead, it seems confused, which indicates a significant lack of coherence in the dialogue. Based on the criteria provided, the response should have continued the thread about 'Inception' and provided insight into the open-ended nature of the movie's conclusion. The AI's response is not relevant and does not demonstrate an understanding of the evolving conversation, which affects its coherence and the insightfulness of the answer.
Rating: [[1]]	

Figure 2: Case of topic shift task from Bai et al. (2024)

6 Additional Ablation

For well-formatted and repetitive input, such as a patient's health log in medical setting, proactive interference only becomes noticeable after a significant number of repetitions. Yet in multi-turn dialogue, PI can appear in merely a few turns of responses. Please see figure2 for reference. We, following Wang and Sun (2025), hypothesize that the memory occupation is determined not only by quantity of context but by the task complexity. Therefore, if we estimate task complexity with the length of answer, the ratio of effect of task complexity shall be examined for better understanding toward model behavior.

7 Cost

The cost estimation can be separated into two parts: encoding the dialogue chunks and additional prompts for CoT. The total #token of the original *LongMemEval_s* is 52 million tokens. If we sample 10 sessions for each QA pair, the expected total #token for our synthesized dialogue can be reduced to around 20.7 million tokens. Regarding the estimation of additional prompts, the total #token may vary greatly to the length of the graph exploration. According to [Li et al. \(2024\)](#), the average #token for the I/O of each question is 52.8k. After comparing the #token in repeated prompts (in the setting of 2 Exploring Atomic Facts, 2 Exploring Chunks, and 1 Exploring Neighbors, the #token is 42.5k per QA) and the given 52.8k, the # output tokens should be around 5k per QA. Given there are 500 QA pairs in the dataset, the expected total #token to be processed would be around 47.1 million tokens, which is \$115 with GPT-4.1 and \$22.5 for GPT-4.1 mini.

[mark for topic-shift aware dialog modeling](#). In *Findings of the Association for Computational Linguistics: EMNLP 2021*, pages 1684–1690, Punta Cana, Dominican Republic. Association for Computational Linguistics.

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